Sichuan University - Pittsburgh Institute

Senior Project Midterm Report

Project Title **Domain-Specific Large Language Model Powered by Retrieval-Augmented Generation for Scholarly Information Retrieval and Understanding**

Major **CS**

	Student ID	Team Member Names
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Introduction

In the contemporary academic landscape, knowledge is expanding at an exponential rate. While this proliferation of scholarly literature fuels scientific progress, it also presents an unprecedented "information overload" challenge for researchers, students, and professionals alike (e.g., Croft, Metzler, & Strohman, 2010). The ability to efficiently discover, comprehend, and synthesize relevant academic information has become a critical bottleneck in research innovation. Traditional keyword-based search engines often falter when handling the semantic nuances of complex research queries; they fail to capture deep context, leading to imprecise results (Guha, McCool, & Miller, 2003; Reimers & Gurevych, 2019). This limitation not only impedes the timely acquisition of knowledge but also intensifies the cognitive burden on researchers during literature analysis and review.

To address this core challenge, this project aims to develop an intelligent platform titled the "Paper Search and Analysis System." Our vision extends beyond traditional "information retrieval" to establish a new paradigm dedicated to "knowledge discovery and synthesis." The system will leverage an advanced hybrid retrieval architecture, integrating the breadth of semantic vector search with the comprehension depth of Large Language Models (LLMs) through Retrieval-Augmented Generation (RAG) technology. The platform will offer users a powerful and intuitive interface, enabling them not only to pinpoint highly relevant literature but also to deeply analyze and interpret paper content

through multi-turn, conversational interaction. Furthermore, the system is designed to streamline and accelerate the entire research workflow. It will specifically address the tedious and error-prone task of manual citation management by providing intelligent citation suggestions and automated formatting. The ultimate goal of this project is to create a robust scholarly tool that significantly enhances research efficiency by providing deeper insights into academic content, thereby empowering scholars to shift their valuable efforts from information gathering to knowledge creation.

Work summary and achievements

During the first phase of the project, our team focused on the critical foundations of the "Paper Search and Analysis System". We have made significant progress in data acquisition, database construction, and the detailed functional design of the frontend architecture. Our achievements are categorized as follows:

1. Database Construction and Data Collection

In accordance with our technical approach to build a local SQLite database for paper metadata, we have successfully completed the initial data collection and ingestion phase.

- **Data Volume:** We have curated and indexed a total of **446 academic papers**, approaching our initial target of a 500-paper library.
- **Domain Distribution:** To ensure the "Domain-Specific" nature of our LLM, we focused heavily on Computer Science disciplines. The distribution of the collected literature is as follows:

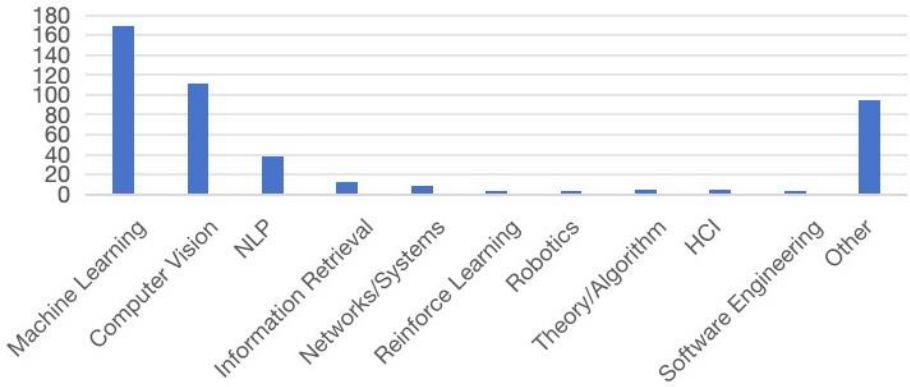

Figure 1. The distribution of the collected literature

2. Frontend Architecture and Functional Module Design

We have finalized the design and logic for the frontend user interface, which utilizes HTML5 and JavaScript. The system has been structured into four distinct functional modules to support the workflow from data ingestion to knowledge retrieval:

- **A. Main Search Interface (Query Module)** To address the limitation of traditional keyword searches, we have designed a dual-mode query interface:
    - **Literal Query (Original Meaning):** Allows users to perform standard searches based on exact semantic matching.
    - **Hierarchical Query (Graded Search):** A novel feature that allows users to tailor the depth of the information retrieved. Users can select from three granularity levels:
        1. **Coarse:** High-level overviews.
        2. **Moderate:** Balanced detail.
        3. **Specific:** In-depth technical details.

Figure 2. Example diagram for paper query

- **B. Paper Upload Interface (Ingestion Module)** To streamline the "Ingestion" pipeline, we designed a semi-automated upload feature:
    - Users can upload citation data in **BibTeX format**.
    - The system automatically parses this data to populate the database fields (Title, Author, Year) and automatically extracts/generates the **Abstract**. This aligns with our goal to automate citation formatting and management.

Figure 3. Example diagram for paper upload

- **C. Library View Interface (Browse Module)**
  - Provides a comprehensive view of the entire database.
  - Includes a **Status Tracking** feature, allowing users to check the "Audit Status" of their uploaded papers, ensuring transparency in the data contribution process.

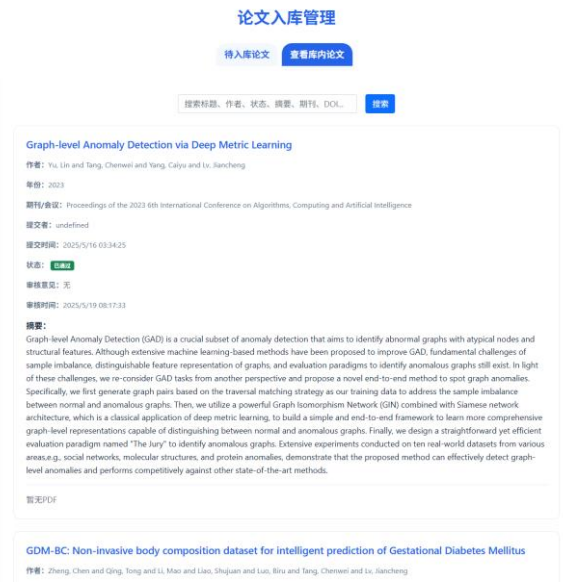

Figure 4. Example diagram for paper management

- **D. Administrator Interface (Management Module)** To ensure data integrity and quality control:
  - **Audit Capability:** Administrators can review user-uploaded papers before they are permanently added to the retrieval index.

- **Database Modification:** Full CRUD (Create, Read, Update, Delete) capabilities to correct metadata or remove irrelevant entries from the SQLite database.

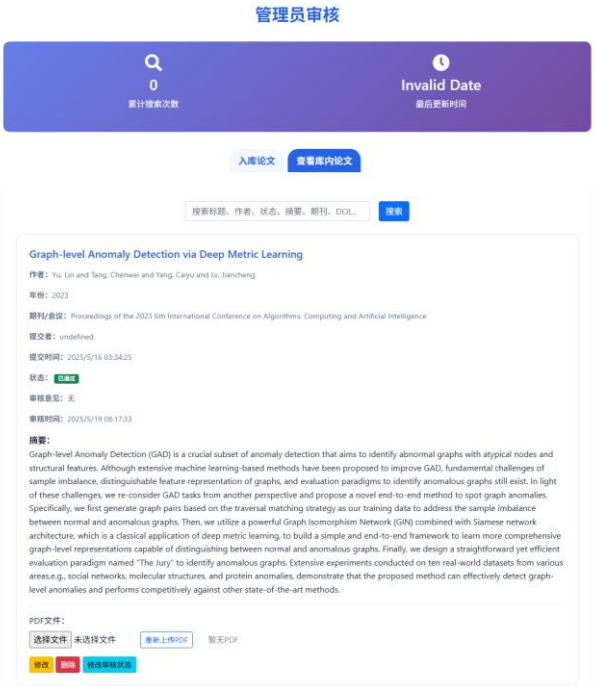

Figure 5. Example diagram for administrator review

3. System Architecture Refinement

We have validated the feasibility of the proposed three-layer architecture (Frontend, Backend, AI Models). The current frontend design effectively interfaces with the planned Flask Orchestrator, preparing the system for the upcoming integration of the RAG pipeline and DeepSeek API.

Individual contribution

1. Xinyu Chen

Role: Project Management

- Progress Tracking: Maintained comprehensive meeting minutes and progress records throughout the development phase, ensuring the team adhered to the estimated timeline.
- Documentation: Responsible for synthesizing technical discussions into formal records, tracking the completion status of project milestones (such as the completion of the database and frontend prototypes), and facilitating communication between team members regarding task dependencies.

2. Kaizhuo Hu

Role: Database Construction

- Data Collection: Led the acquisition and curation of the domain-specific academic literature. Successfully collected and indexed 446 papers to build the core dataset.
- Domain Classification: Organized the dataset to ensure high relevance for the CS domain model, ensuring a distribution covering Machine Learning (169), Computer Vision (111), NLP (38), and other key areas like Networks and Robotics.
- Database Management: Responsible for the initial setup of the data ingestion pipeline, ensuring the collected PDF and metadata align with the

database schema (Title, Author, Abstract, etc.).

3. Jian Yuan

Role: Frontend Development & System Interaction Design

- Frontend Architecture: Responsible for the complete design and implementation of the web-based user interface using HTML5 and JavaScript .
- Module Implementation: Successfully developed four key functional modules:
    - Query Interface: Implemented the dual-mode search logic, including "Literal Query" for exact matching and "Hierarchical Query" (Coarse, Moderate, Specific) for graded information retrieval.
    - Upload Interface: Developed the logic for the "Upload" page, including the automated BibTeX parsing feature that extracts metadata and populates abstracts.
    - Management Interfaces: Built the "Library View" for database browsing and the "Admin Dashboard" for auditing and modifying paper data.

The proposed timeline from the proposal and schedule attainment

1. Overview of Proposed Timeline

The project schedule for the Fall Semester was designed to progress from feasibility analysis to technical implementation. As detailed in the proposal, the critical development phase was scheduled for Weeks 11-13.

2. Schedule Attainment Analysis(Fall Semester)

We have strictly adhered to the proposed timeline and, in the technical implementation phase (Weeks 11-13), we have exceeded the initial scope by completing the frontend interface ahead of the integration phase.

Timeframe	Proposed Milestone (from Proposal)	Actual Status & Achievement	Attainment
Weeks 1-2	Meet with the supervisor and classmates to discuss the advanced project tasks	**Completed.** Defined project scope and team roles.	100%
Weeks 2-3	Discuss the project topic and determine the feasibility of implementation	**Completed.** Feasibility confirmed.	100%
Weeks 4-7	Determine the proposal structure and complete the proposal writing by the 8th week	**Completed.** Proposal drafted and finalized.	100%
Week 8	Submit the proposal and prepare for the defense	**Completed.** Proposal submitted and approved.	100%
Weeks 9-10	Conduct literature review and technology selection	**Completed.** Finalized the tech stack: Flask (Backend), SQLite (DB), and DeepSeek API (LLM).	100%
Weeks 11-13	Prepare and adjust the model, preprocess the data and modify the deep learning code	**Completed & Advanced.** We significantly expanded the scope of "modify code" and "data preprocessing": 1. **Data Collection:** Successfully collected and preprocessed **446 domain-specific papers** (ML, CV, NLP, etc.). 2. **Frontend Development:** Fully	100% (Ahead)

		implemented the user interface code, including the **Main Search** (Hierarchical Query), **Upload**, **Library**, and **Admin** modules.	
Weeks 14-18	Start writing the interim report	**In Progress.** Currently summarizing the database statistics and frontend architecture.	100%
Week 19	Prepare the interim defense	**Upcoming.** Preparing the live demonstration of the frontend and database.	On Track

3. Spring Semester

Weeks 1-2	Complete the development and integration of core functions
Weeks 3	Perform System Acceptance Testing (SAT); Start writing the final report and research paper
Weeks 4	Conduct User Acceptance Testing (UAT); Analyze test results for the report
Week 5-9	Discuss and revise the report and paper within the group
Weeks 9	Submit the report
Weeks 10-11	Revise the report
Weeks 12-13	Prepare for the final defense
Weeks 14-16	Submit all materials

References

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Advisor Signature _____ **Date** _____